

CAS Machine Learning – Time Series

Welcome

Dr. Ludovic Amruthalingam
ludovic.amruthalingam@hslu.ch

Informatik



About me

Studied computer science at EPFL & ETH

PhD in AI for clinical dermatology at the University of Basel

- AI decision support
- AI challenges & opportunities for dermatology clinics

Dozent at HSLU

- 80% research projects with industry in MedTech & FinTech
- 20% lecturing: Time Series Analysis, Explainable AI, Medical Imaging



AAI
Research Lab

derma²go
Ihre Experten für gesunde Haut

derec

Helmuth Fischer
STIFTUNG

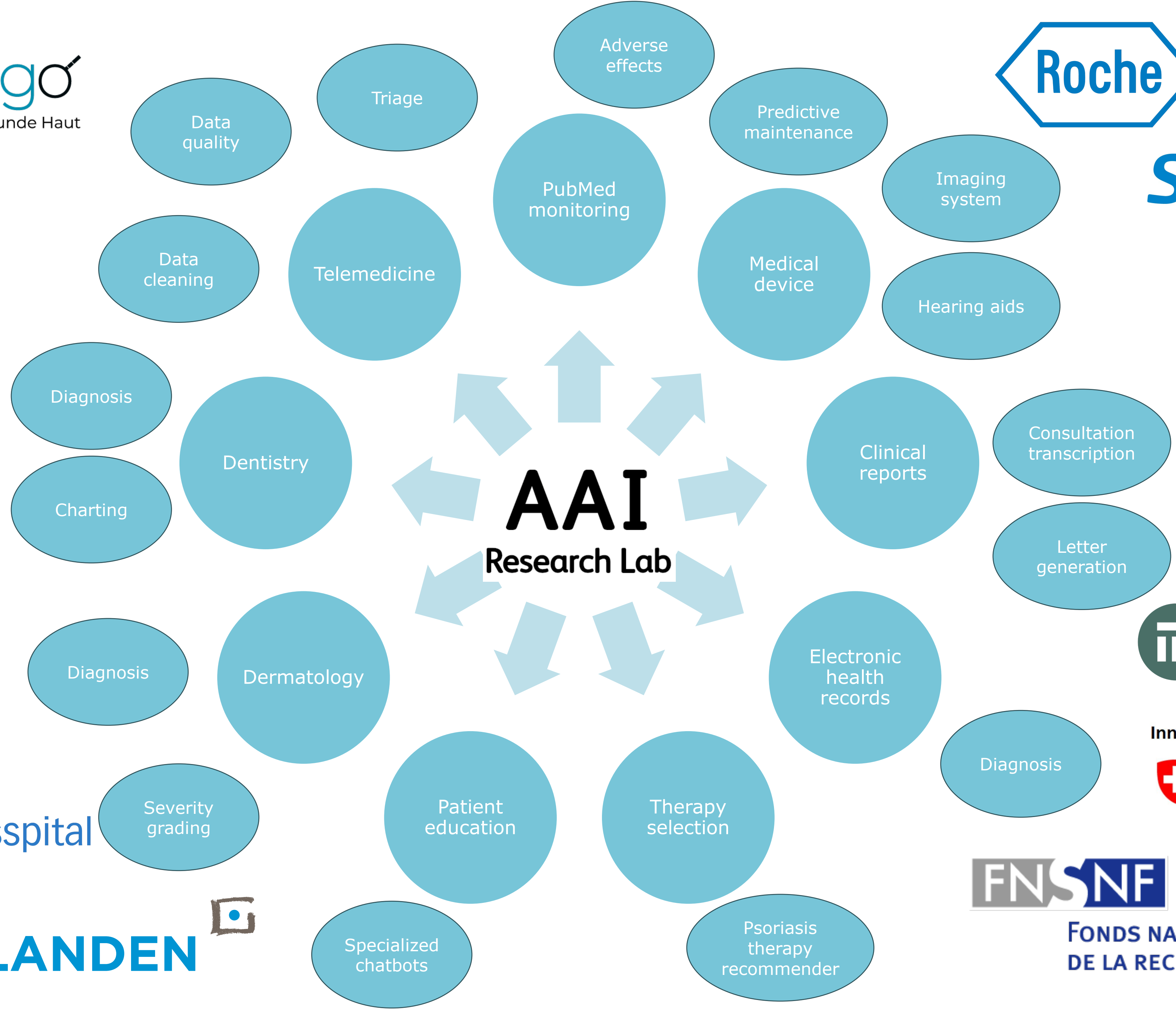
VELUX
STIFTUNG

Universitätsspital
Basel

luzerner kantonsspital

HSLU

HIRSLANDEN



Roche

sonova
HEAR THE WORLD



I.O.M. HEALTH
Institut Osteopathische Medizin

oerli
SWITZERLAND

Swiss Re

Innosuisse - Swiss Innovation Agency

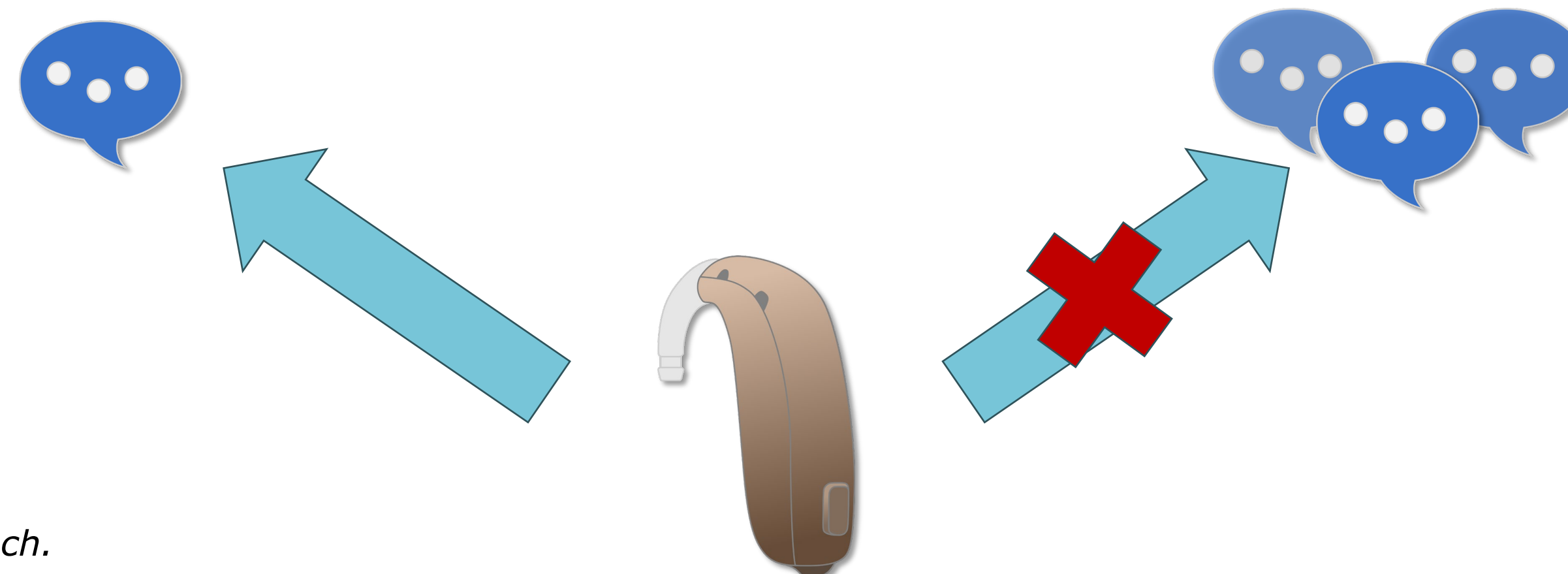


Schweizerische Eidgenossenschaft
Confédération suisse
Confederazione Svizzera
Confederaziun svizra

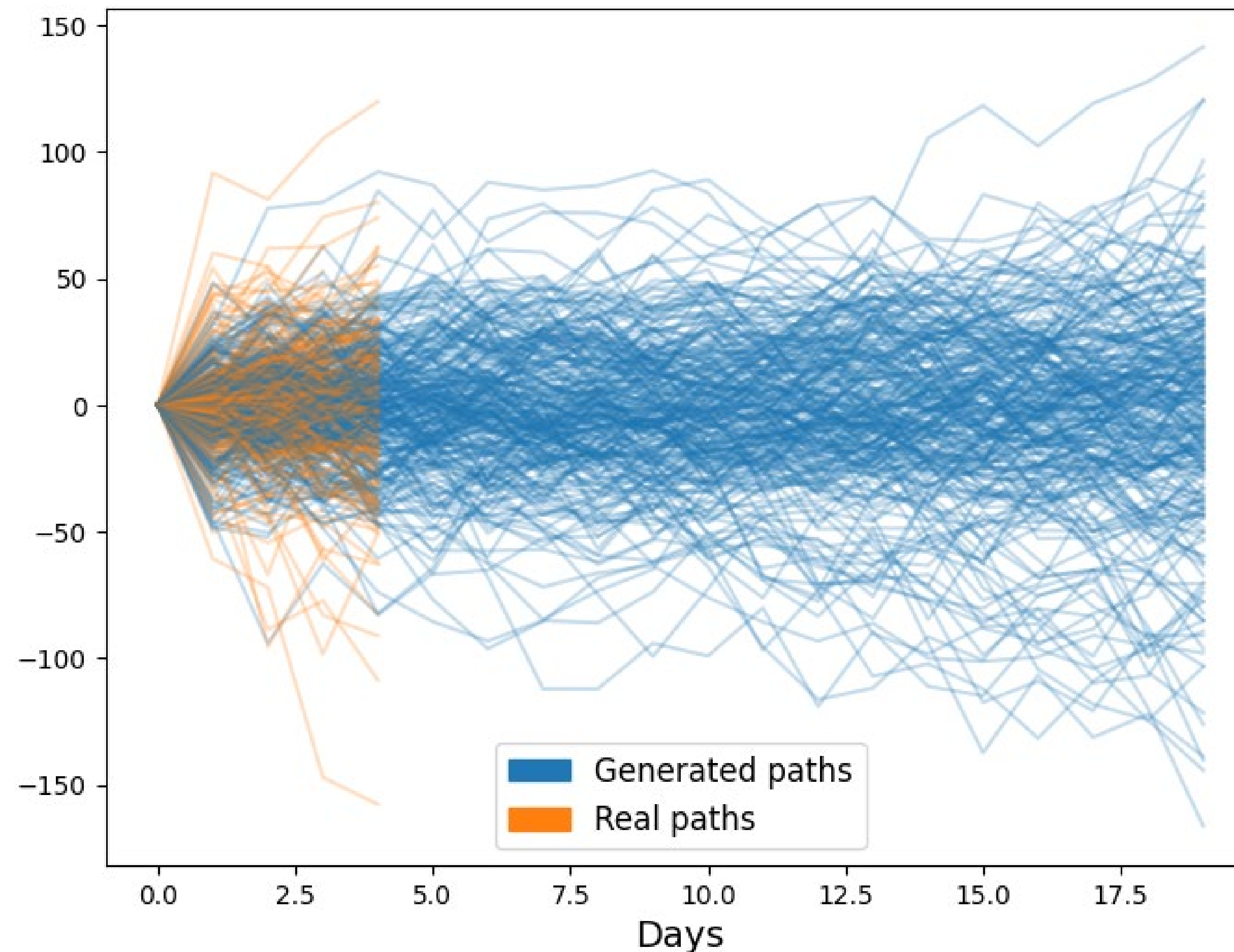
FNSNF

FONDS NATIONAL SUISSE
DE LA RECHERCHE SCIENTIFIQUE

Hearable steering in complex acoustic scenes



Benchmarking Synthetic Time Series Quality



Agenda for today

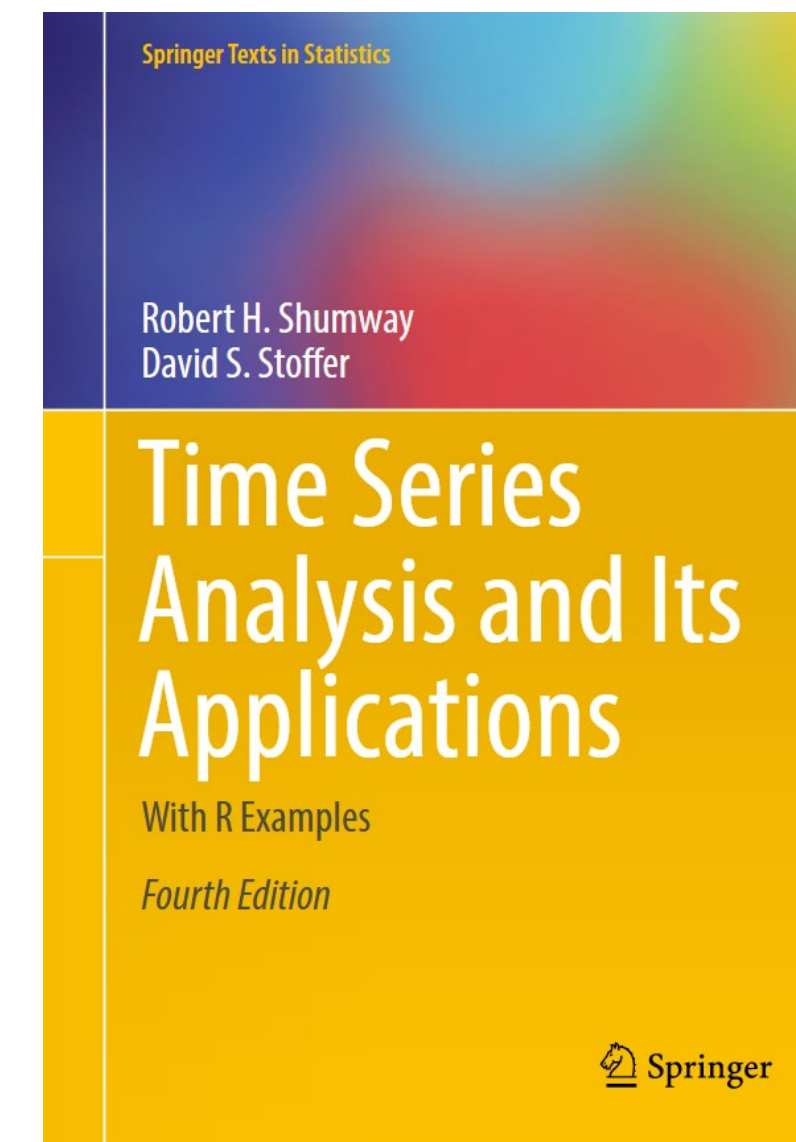
Core time series concepts

Classical time series models: forecasting with ARIMA

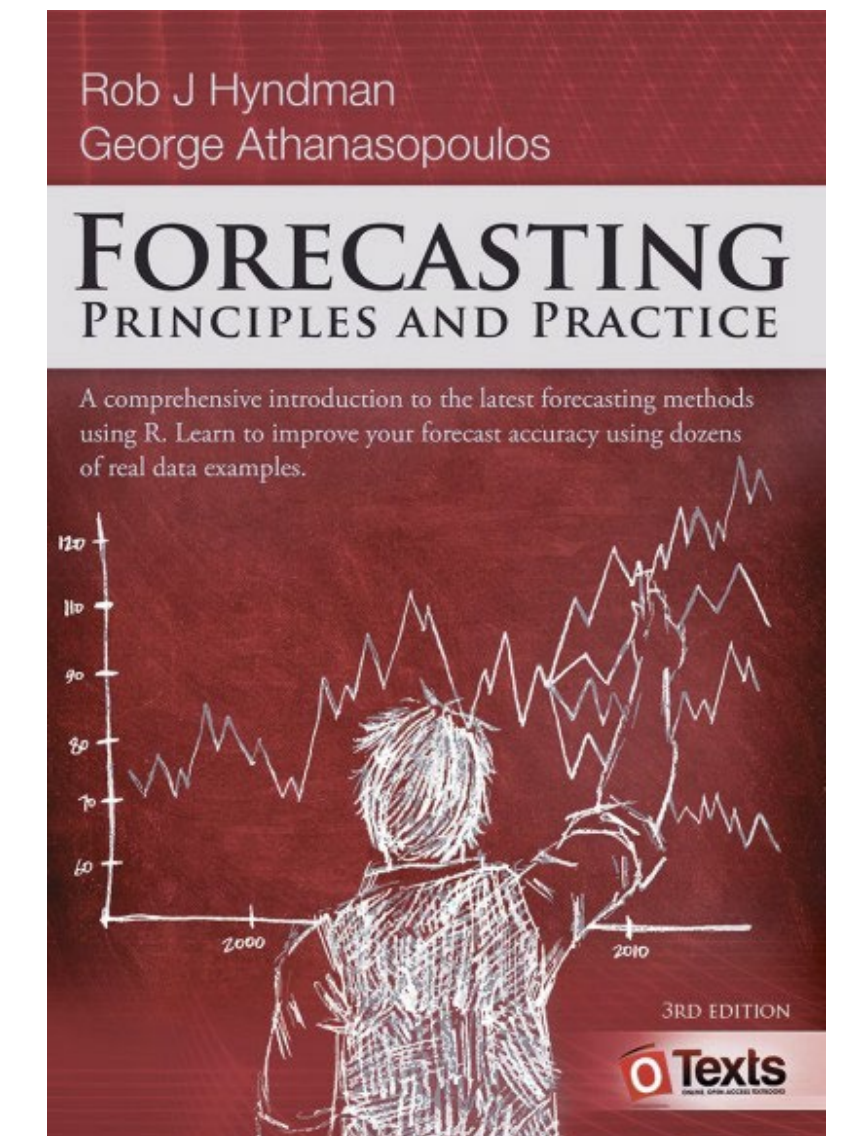
Machine learning and deep learning for time series

Literature

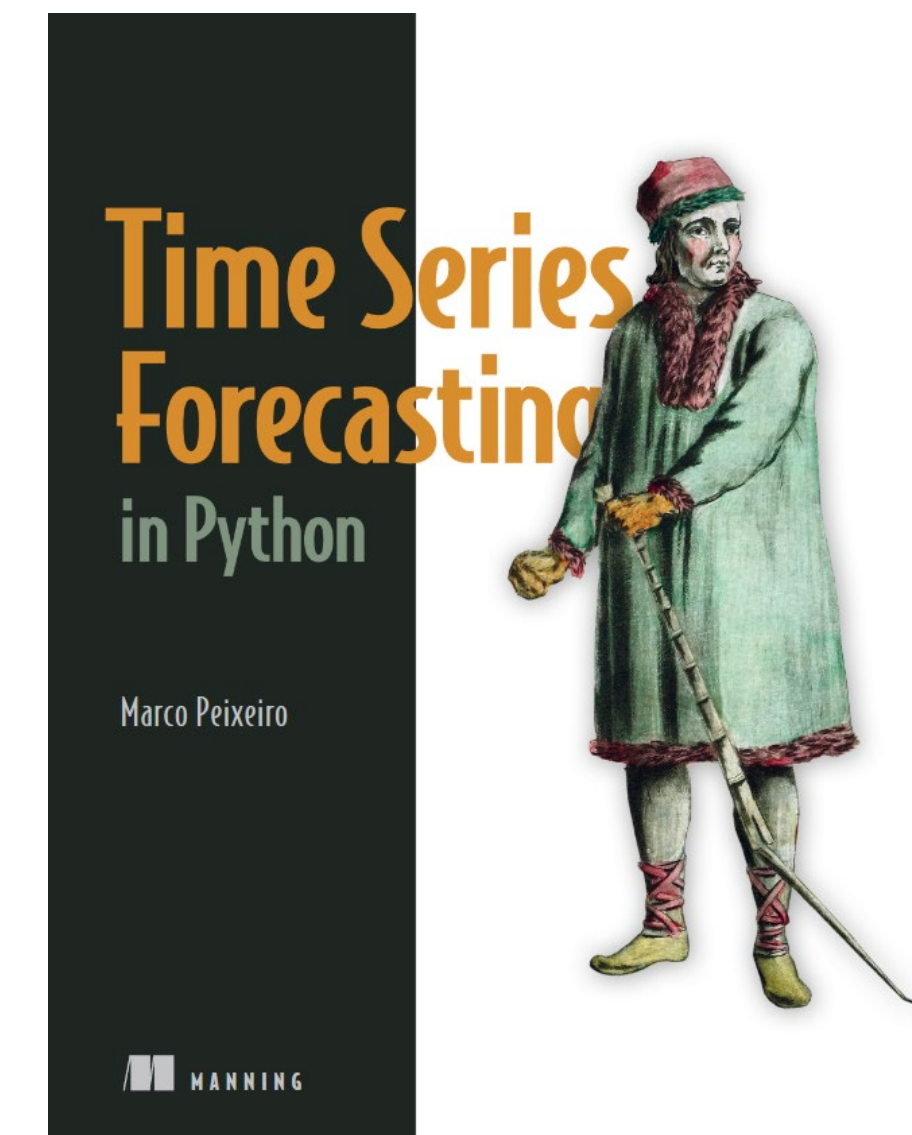
Time Series Analysis and its Applications, Shumway & Stoffer, Springer, 4th edition (2017).



Forecasting Principles and Practice, Hyndman & Athanasopoulos, Otexts, 3rd edition (2021).



Time Series Forecasting in Python, Peixeiro, Manning, 1st edition, (2022).



Warm-up

Where do you encounter time-based data?